

Customer Sign-Up Behaviour & Data Quality Audit

Business Intelligence Project | Rapid Scale SaaS

Prepared for Marketing & Onboarding decision-making

1. Introduction

This report analyses customer sign-up data to identify patterns in acquisition, customer demographics, marketing preferences and subscription-plan selection. It also assesses data quality and completeness so that business decisions are based on reliable information.

The dataset initially contained 516 customer records across 10 fields, including customer details, signup dates, acquisition source, region, subscription plan, marketing preference, age and gender. The data was reviewed and cleaned before analysis, with particular attention to duplicate customers, invalid dates, inconsistent categories, malformed email addresses, missing values and implausible ages.

The analysis examined weekly signup activity, acquisition sources, regional completeness, marketing opt-in behaviour and plan preferences by age group. The results are used to answer the four required business questions and provide practical recommendations.

2. Data Cleaning Summary

Cleaning decisions were designed to improve consistency while avoiding unsupported assumptions. Records were retained where the available information remained useful and a reliable replacement value could not be determined.

Field / issue	Extent	Action taken	Business rationale
Signup dates	2 invalid; 2 future-dated	Converted to datetime; invalid values became NaT; future dates flagged	Prevents unreliable time-based analysis without discarding useful customer records.
Duplicate customers	16 duplicate rows	Retained the most recent record for each customer_id	Prevents customer counts and segment analysis being inflated.
Plan values	Multiple inconsistencies	Trimmed spaces, standardised capitalisation and corrected "Premuim" to "Premium"	Ensures the same plan is not split into multiple categories.
Gender values	Multiple inconsistencies	Trimmed spaces and standardised capitalisation	Ensures consistent demographic grouping.
Email	20 missing; 15 malformed	Missing retained; whitespace removed and unambiguous "_at_" entries corrected	Avoids inventing missing contact details while restoring clearly intended addresses.
Region	60 missing (12.0%)	Missing values retained	Region could not be reliably inferred from available fields.
Age	25 missing after cleaning	Values below 18 or above 100 converted to missing	Avoids unsupported age estimates and reduces

			distortion from implausible values.
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After cleaning, ages ranged from 18 to 65 years, with a mean of 33.32 and median of 32. There were 25 missing ages (5.0%). Age groups used for segmentation were 18–24, 25–34, 35–44, 45–54 and 55–65.

Future signup dates were flagged rather than automatically removed. Two records had invalid signup dates and therefore could not be assigned reliably to a week or month.

3. Key Findings & Trends

3.1 Signup activity

Weekly signup activity was very low between 2019 and 2023 and increased substantially from 2024 onwards, with more frequent signups throughout 2025 and 2026. The highest observed weekly total was 10 signups. Of the 500 cleaned customer records, 498 had valid signup dates and were included in weekly analysis.

3.2 Acquisition sources

Google was the largest acquisition source overall, with 162 customers (32.4%), followed by Instagram with 110 (22.0%) and Referral with 67 (13.4%). Google and Instagram together accounted for 54.4% of all customers. This is descriptive evidence only; acquisition investment decisions would require conversion, cost and customer-value measures.

The latest month with recorded non-future signup activity was August 2026, when one signup was recorded and the source was Google. Because this is a single observation, it should not be treated as representative of normal monthly acquisition performance.

3.3 Marketing opt-in and age

Marketing opt-in rates generally decreased across the age groups: 68.42% for 18–24, 52.87% for 25–34, 39.68% for 35–44, 30.00% for 45–54 and 38.89% for 55–65. The pattern suggests that younger customers were generally more likely to opt into marketing, although the relationship was not perfectly linear.

4. Business Question Answers

4.1 Which acquisition source brought in the most users last month?

Google was the source of the only recorded signup in August 2026, the latest month with recorded non-future signup activity. This result has limited significance because only one signup was recorded. Acquisition performance should therefore be assessed across several complete months.

4.2 Which region shows signs of missing or incomplete data?

Asia-Pacific shows the highest proportion of incomplete customer records among the known regions. Of its 90 customer records, 10 had at least one missing value in email, signup date or age, representing 11.11% of the region. North America had more affected records in absolute terms (15), but its proportion was lower at 10.27%. Separately, region itself is missing for 60 customers (12.0% of the full dataset), so regional comparisons should be interpreted with caution.

4.3 Are older users more or less likely to opt into marketing?

Older customers were generally less likely to opt into marketing than younger customers. The highest opt-in rate was 68.42% among 18–24-year-olds, compared with 30.00% among 45–54-year-olds. The 55–65 group was slightly higher at 38.89%. These results indicate an association between age group and observed opt-in behaviour, rather than demonstrating that age directly causes the difference.

4.4 Which plan is most commonly selected, and by which age group?

Basic was the most common plan overall, selected by 240 customers (48.0%), followed by Pro at 184 (36.8%) and Premium at 76 (15.2%). Basic was also the most common plan among customers aged 18–24, 25–34, 35–44 and 55–65. The exception was the 45–54 group, where Pro was most common with 23 customers, compared with 20 selecting Basic and 17 selecting Premium.

5. Recommendations

Improve data completeness. Introduce stronger validation at customer registration for key fields such as region, email and age. Mandatory or validated fields and automated quality checks would reduce incomplete records entering downstream reporting.

Monitor acquisition performance over time. Google is the largest acquisition source overall, but the latest monthly observation contains only one signup. Track several complete months and combine signup volume with conversion, campaign cost and customer value before changing marketing investment.

Test age-based marketing strategies. Use the observed differences in opt-in rates to test age-appropriate messaging, channels or incentives, particularly for older customer groups. Testing should be used rather than assuming a particular intervention will work.

Investigate plan preferences by age. Basic dominates overall, while Pro leads among customers aged 45–54. Further analysis of plan features, pricing and customer value could establish whether this group has distinct needs or purchasing behaviour.

6. Data Issues or Risks

The most significant data-quality risk is incomplete regional information. Region is missing for 60 of the 500 cleaned customer records (12.0%), meaning a substantial portion of customers cannot be confidently assigned to a geographic segment. This can weaken regional campaign reporting and make differences between regions harder to interpret.

A second important limitation concerns the latest-month acquisition result. August 2026 contains only one recorded non-future signup, so Google being the leading source in that month should not be interpreted as a reliable trend. Future signup dates were retained and flagged for review, and 25 customers have missing ages.

At source, the business should strengthen validation rules, monitor missingness routinely and investigate why incomplete fields occur. Future reporting should also distinguish incomplete periods from complete periods so that partial-month activity is not mistaken for a normal performance level.

Conclusion

The cleaned dataset shows clear growth in signup activity from 2024 onwards, with Google the largest acquisition source overall. Younger customers generally show higher marketing opt-in rates, while Basic is the dominant subscription plan across most age groups. The analysis also identifies meaningful data-quality risks, particularly missing regional information. Combining stronger data-entry controls with longer-term acquisition and customer-segmentation monitoring would improve the reliability and usefulness of future business reporting.

Analysis tools: Python, Pandas, Matplotlib, Google Colab